Black-box variational inference (BBVI)

Introduced by Ranganath, Gerrish, and Blei (2014) and recently won the "Test of Time" award at AISTATS 2014. This is a general recipe for gradient-based variational inference in a wide range of (non-conjugate) models.

Some notation: Define the "learning signal" g(z, lam) and the ELBO as its expectation:

$$g(z,\lambda) \equiv \log \frac{p(x,z)}{q(z|\lambda)} \qquad \mathcal{L}(\lambda) \equiv \mathbb{E}_{z\sim q_\lambda}[g(z,\lambda)]$$

BBVI is based on the "score function trick". Consider the gradient of the ELBO:

$$\nabla_\lambda \mathcal{L}(\lambda) = \nabla_\lambda \mathbb{E}_{q_\lambda}[g(z,\lambda)]$$

$$= \nabla_\lambda \int q(z|\lambda)\, g(z,\lambda)\, dz$$

In this setting we can swap the gradient and integral:

$$= \int \nabla_\lambda q(z|\lambda)\, g(z,\lambda)\, dz$$

To reduce clutter we will just say "g" and "q". This breaks into two terms:

$$= \underbrace{\int g(\dots)\nabla q(\dots)}_{1} + \underbrace{\int q(\dots)\nabla g(\dots)}_{2}$$

Working on the first term first, we will use the following identity: $\nabla \log q = \frac{\nabla q}{q}$

$$= \int g(\dots)\, q(\dots)\nabla \log q(\dots) \; + \; 2$$

$$= \mathbb{E}_q[g(\dots)\nabla \log q(\dots)] \; + \; 2$$

Working on the second term, we expand g(...) and notice that p(...) is const wrt lambda:

$$= 1 + \int q(\dots)\,\nabla_\lambda(\log p(\dots) - \log q(\dots))$$

$$= 1 + \int q(\dots)\,\nabla \log q(\dots)$$

Apply the same identity, as above we find that q(...) cancels. The gradient and integral can be swapped again, and we find that the second term is 0

$$= 1 - \int \cancel{q(\dots)}\,\frac{\nabla q(\dots)}{\cancel{q(\dots)}}$$

$$= 1 - \underbrace{\nabla \underbrace{\int q(z|\lambda)\,dz}_{=1}}_{=0}$$

In conclusion, the gradient of an expectation is the expectation of a gradient:

$$\nabla_\lambda \mathbb{E}_q[g(z,\lambda)] = \mathbb{E}_q[g(z,\lambda)\nabla \log q(z|\lambda)]$$

This then leads naturally to the "score function gradient estimator":

$$\hat{\nabla}_\lambda \mathcal{L}(\lambda) = \frac{1}{S}\sum_{s=1}^{S} g(z_s, \lambda)\, \nabla_\lambda \log q(z_s \mid \lambda)$$

$$z_s \overset{iid}{\sim} q(z \mid \lambda) \quad s = 1 \dots S$$

This same estimator was previously introduced for reinforcement learning in the 1990s and goes by the name REINFORCE.

The vanilla "black box variational inference (BBVI)" algorithm is then:

$$\boxed{\begin{array}{l} \text{for } t = 1, 2, \dots \text{ until convergence:} \\ \qquad z_s \overset{iid}{\sim} q(z \mid \lambda) \\ \qquad \lambda = \lambda + \rho_t \hat{\nabla} \mathcal{L}(\lambda) \end{array}}$$

...which is guaranteed to converge to a local mode of the ELBO provided that the step-sizes follow the Robins-Monro conditions. Notice how the only requirements are:

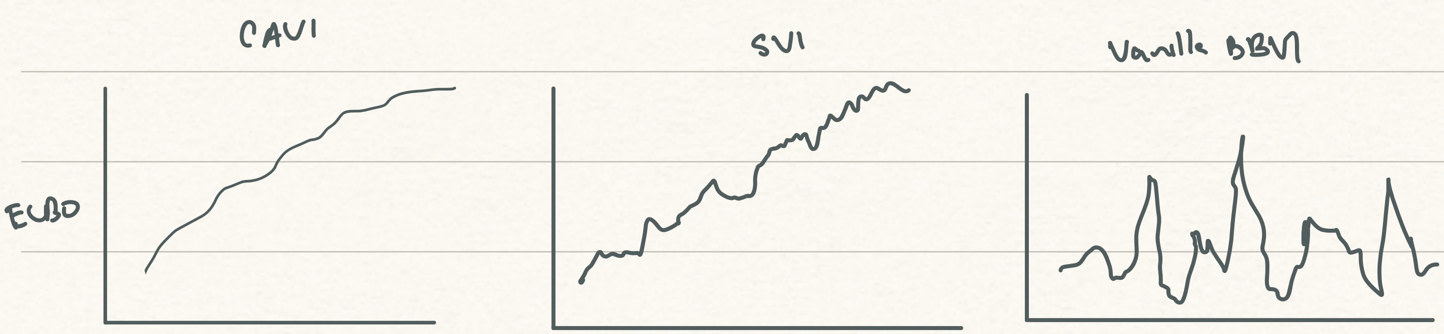
- We can evaluate the density $p(x, z)$
- We can evaluate the density $q(z \mid \lambda)$
- We can evaluate the gradient $\nabla_\lambda q(z \mid \lambda)$
- We can sample $z \sim q(z \mid \lambda)$

Dealing with gradient variance

There are few settings in which vanilla BBVI actually works. The reason is that although the "score function gradient estimator" is unbiased, it's variance is typically too high:

$$\mathbb{E}[\hat{\nabla}\mathcal{L}(\lambda)] = \nabla\mathcal{L}(\nabla) \checkmark \qquad \mathrm{Var}(\hat{\nabla}\mathcal{L}(\lambda))$$ 😱

With CAVI the ELBO monotonically increases every iteration. With SVI the ELBO isn't guaranteed to increase anymore (due to stochastic subsampling), but moving averages of small window sizes still often go up in practice. With vanilla BBVI, the gradient variance is often so high that no progress occurs within acceptable time horizons:



Rao-Blackwellization

The first strategy to deal with this is called Rao-Blackwellization. This takes advantage of conditional independencies in the model p(...) and variational distribution q(...). Assume that we are making the mean-field assumption:

$$\log q(z|\lambda) = \sum_d \log q(z_d|\lambda_d) \equiv \sum_d \log q_d$$

For a given latent variable z_d, the log joint distribution can always be written as:

$$\log p(z,x) = \log p_d(z_{(d)}, x) + \log p_{\setminus d}(z_{\setminus d}, x)$$

where the first term are all the terms in the log joint that involve z_d and the second is all the terms in the log joint that don't. The first term is thus a function of z_d and the Markov blanket of z_d:

$$z_{(d)} \equiv z_d \cup \{ z_{d'} : z_{d'} \in \underbrace{\text{Child}(z_d) \cup \text{parent}(z_d) \cup \text{co-parent}(z_d)}_{\equiv \text{MarkovBlanket}(z_d)} \}$$

To reduce notation we will refer to the terms as just p_d and p_\d (but NOTE they are not probability distributions, unlike q_d):

$$\log p(z,x) = \log p_d + \log p_{\setminus d}$$

Now consider a single component of the gradient:

$$\nabla_\lambda \mathcal{L}(\lambda) \equiv \begin{bmatrix} \vdots \\ \nabla_{\lambda_d} \mathcal{L}(\lambda) \\ \vdots \end{bmatrix} \qquad \nabla_{\lambda_d} \mathcal{L}(\lambda) = \mathbb{E}_q\left[\log \frac{p(x,z)}{q(z|\lambda)} \nabla_{\lambda_d} \log q(z|\lambda)\right]$$

This breaks into two terms where note that: $\nabla_{\lambda_d} \log q(z|\lambda) = \nabla_{\lambda_d} q_d, \quad q_{\setminus d} \equiv \prod_{j \neq d} q_j$

$$= \mathbb{E}_q\left[\log \frac{p_d}{q_d} \nabla_{\lambda_d} \log q_d\right] + \mathbb{E}_q\left[\log \frac{p_{\setminus d}}{q_{\setminus d}} \nabla_{\lambda_d} \log q_d\right]$$

By law of total expectation, we can factorize E_q and then push E_qd in:

$$= \mathbb{E}_q\left[\log \frac{p_d}{q_d} \nabla_{\lambda_d} \log q_d\right] + \mathbb{E}_{q_{\setminus d}}\left[\log \frac{p_{\setminus d}}{q_{\setminus d}} \underbrace{\mathbb{E}_{q_d}\left[\nabla_{\lambda_d} \log q_d\right]}_{=0}\right]$$

Since the expectation of the score function is 0, the second term drops. Meanwhile the first term can be written as an expectation over only the Markov blanket:

$$= \mathbb{E}_{q_{(d)}}\left[\log \frac{p_d}{q_d} \nabla_{\lambda_d} \log q_d\right], \quad q_{(d)} \equiv q(z_{(d)}|\lambda)$$

So the Rao-Blackwellized gradient estimator is then:

$$\hat{\nabla}^{RB}_{\lambda_d} = \frac{1}{S} \sum \log \frac{p_d(z^s_{(d)}, x)}{q(z_d \mid \lambda_d)} \nabla_{\lambda_d} \log q(z_d \mid \lambda_d)$$

Intuitively, you might think this will have lower variance since each estimate involves sampling fewer latent variables (only the Markov blanket). Indeed we can prove it is a lower variance estimator by showing that this is an instance of Rao-Blackwellization.

Say we want to estimate the expectation of a function of two variables:

$$\mathbb{E}_{x,y \sim p(x,y)} [J(x,y)]$$

And say we also can calculate exactly the following conditional expectation function:

$$\hat{J}(x) \equiv \mathbb{E}_{y \sim p(y \mid x)} [J(x,y) \mid x]$$

By the law of total variance, the variance of the joint function equals:

$$\mathrm{Var}(J(x,y)) = \underbrace{\mathrm{Var}(\mathbb{E}[J(x,y) \mid x])}_{= \mathrm{Var}(\hat{J}(x))} + \underbrace{\mathbb{E}[\mathrm{Var}(J(x,y) \mid x)]}_{\geq 0}$$

which is therefore greater than the variance of the conditional expectation function:

$$\geq \mathrm{Var}(\hat{J}(x))$$

We can see that that Rao-Blackwellized gradient estimator above conforms to this pattern. At a high-level what we did was rewrite the expectation over all z as:

$$\mathbb{E}_z[\cdots] = \mathbb{E}_{z_{\setminus d}}[\mathbb{E}_{z_d}[\cdots \mid z_{\setminus d}]]$$

and then we used the fact that (parts of) the inner expectation could be computed exactly.

Control variates

The other standard approach to decreasing gradient variance is via "baselines" or "control variates". We will cover two of such cases often used for BBVI.

Notice that since the gradient of the score function is zero, the following holds:

$$\mathbb{E}[g(\cdots)\nabla \log q(\cdots)] = \mathbb{E}[(g(\cdots) - b)\nabla \log q(\cdots)]$$

for any constant b. This motivates the introduction of a "baseline", which is a constant b such that a Monte Carlo estimate of the RHS has lower variance than a Monte Carlo estimate of the LHS. One such constant that may intuitively provide this property is the mean of the learning signal. Define the centered learning signal:

$$\tilde{g}(z,\lambda) \equiv g(z,\lambda) - \mathbb{E}[g(z,\lambda)]$$

Notice that the expected value of the learning signal is simply the ELBO. We do not know the ELBO in a closed form, however we can estimate it. This motivates the "**leave-one-out (LOO) score function gradient estimator**":

$$\hat{\nabla}^{\text{loo}}\mathcal{L}(\lambda) = \frac{1}{S}\sum_{s=1}^{S}\left(g(\lambda, z_s) - \frac{1}{S-1}\sum_{s'\neq s} g(\lambda, z_{s'})\right)\nabla_\lambda \log q(z_s|\lambda)$$

$$z_s \overset{\text{iid}}{\sim} q(z|\lambda) \qquad s=1\ldots S$$

You might worry that the samples being reused to estimate the baseline will bias the estimator. However notice that since they are drawn iid, and since we are leaving-one-out to sample, the estimator is unbiased:

$$\mathbb{E}_{z_1\cdots z_S}\left[\hat{\nabla}^{\text{loo}}\mathcal{L}(\lambda)\right] = \frac{1}{S}\sum_{S=1}^{S}\mathbb{E}_{z_s}[g(\lambda,z_s)\nabla_\lambda \log q(z_s|\lambda)]$$

$$- \frac{1}{S}\frac{1}{S-1}\sum_s\sum_{s'\neq s}\mathbb{E}_{z_{s'}}[g(\lambda,z_{s'})]\,\mathbb{E}_{z_s}\underbrace{[\nabla_\lambda \log q(z_s|\lambda)]}_{=0}$$

More generally, a "control variate" with respect to some function f(z) is a family of functions defined by the following:

$$\hat{f}(z) = f(z) - a\,(h(z) - \mathbb{E}[h(z)])$$

Notice that for any a and h(z), the following holds:

$$\mathbb{E}[\hat{f}(z)] = \mathbb{E}[f(z)]$$

It then makes sense to find values of h(z) and a such that we reduce the variance:

$$\text{Find } \{h(z), a\} \text{ s.t. } \mathrm{Var}(\hat{f}) < \mathrm{Var}(f)$$

The variance of the control variate equals:

$$\mathrm{Var}(\hat{f}) = \mathrm{Var}(f) + a^2\,\mathrm{Var}(h) - 2a\,\mathrm{Cov}(f,h)$$

So intuitively a good choice of h(z) is one with high covariance with f(z). The original BBVI paper advocates for the score function:

$$f(z) \equiv g(z,x)\,\nabla \log q(z|x)\,, \qquad h(z) \equiv \nabla \log q(z|x)$$

For a given h(z) we can then find the best a by taking the gradient wrt the variance:

$$\frac{\partial}{\partial a}\mathrm{Var}(\hat{f}) = 2a\,\mathrm{Var}(h) - 2\,\mathrm{cov}(f,h)$$

Setting this equal to zero results in the optimal a:

$$\underset{a}{\mathrm{argmin}}\;\mathrm{Var}(\hat{f}) = \frac{\mathrm{Cov}(f,h)}{\mathrm{Var}(h)}$$

The original BBVI paper thus recommends, estimating the Cov(f, h) and Var(f) terms using the Monte Carlo samples to compute an estimate of the optimal value of a. For more details, see the paper.